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### Systems and methods for controlling additive manufacturing

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#### Abstract

Methods and associated systems and apparatus for determining a tool path for use in additively manufacturing a structure. The methods may include receiving data that at least partially defines an outer boundary for a region of the structure, partitioning the region based on placement of a partitioning line across the region from the outer boundary, and generating the tool path to navigate at least part of the outer boundary and to cross, at least partially, the region based on the placement of the partitioning line.

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## **Background/Summary**

RELATED APPLICATION (1) This application is based on and claims the benefit of priority from U.S. Provisional Application No. 63/200,250 that was filed on Feb. 24, 2021, the contents of which are expressly incorporated herein by reference.

### **TECHNICAL FIELD**

(1) The present disclosure relates generally to manufacturing control systems and, more particularly, to systems and methods for controlling additive manufacturing.

### **BACKGROUND**

(2) Additive manufacturing is a process of creating three-dimensional parts by depositing overlapping layers of material under the guided control of a computer. A common form of additive manufacturing is known as fused deposition modeling (FDM). Using FDM, a thermoplastic is passed into and liquified within a heated print head. The print head is moved in a predefined trajectory (a.k.a., a tool path) as the material discharges from the print head, such that the material is laid down in a particular pattern and shape of overlapping 2-dimensional layers. The material cools after exiting the print head and hardens into a final form. A strength of the final form is primarily due to properties of the thermoplastic supplied to the print head and a 3-dimensional shape formed by the stack of 2-dimensional layers.

(3) A recently developed improvement over FDM manufacturing involves the use of continuous reinforcements, which in some examples include fibers embedded within material discharging from the print head. For example, a matrix is supplied to the print head and discharged (e.g., extruded and/or pultruded) along with one or more continuous fibers also passing through the same head at the same time. The matrix can be a traditional thermoplastic, a powdered metal, a liquid matrix (e.g., a snap-curing and/or two-part resin), or a combination of any of these and other known matrixes. Upon exiting the print head, a cure enhancer (e.g., a UV light, a laser, an ultrasonic emitter, a heat source, a catalyst supply, etc.) is activated to initiate and/or complete curing (e.g., hardening, cross-linking, polymerization, sintering, etc.) of the matrix. This curing occurs almost immediately, allowing for unsupported structures to be fabricated in free space. And when fibers, particularly continuous fibers, are embedded within the structure, a strength of the structure may be multiplied beyond the matrix-dependent strength. An example of this technology is disclosed in U.S. Pat. No. 9,511,543 that issued to Tyler on Dec. 6, 2016 ("the '543 patent").

(4) The disclosed systems and methods are directed to addressing ways of controlling additive manufacturing systems similar to those disclosed in the '543 patent and/or other systems known in the art.

### **SUMMARY**

(5) According to an aspect, there is provided a method of determining a tool path for use in additively manufacturing a structure. The method may include receiving data that at least partially defines an outer boundary for a region of the structure and partitioning the region based on placement of a partitioning line across the region from the outer boundary. The method may also

include generating the tool path to navigate at least part of the outer boundary and to cross, at least partially, the region based on the placement of the partitioning line.

(6) According to an aspect, there is provided a computer program product comprising computer program code that, when executed on a computer processor, is configured to perform a method. The method may include receiving data that at least partially defines an outer boundary for a region of the structure and partitioning the region based on placement of a partitioning line across the region from the outer boundary. The method may also include generating the tool path to navigate at least part of the outer boundary and to cross, at least partially, the region based on the placement of the partitioning line.

(7) According to an aspect, there is provided an additive manufacture system, including a moveable support, a print head connected to the moveable support and configured to discharge a material, and a computer processor. The computer processor may include computer program code that, when executed on the computer processor, is configured to perform a method. The method may include receiving data that at least partially defines an outer boundary for a region of the structure and partitioning the region based on placement of a partitioning line across the region from the outer boundary. The method may also include generating the tool path to navigate at least part of the outer boundary and to cross, at least partially, the region based on the placement of the partitioning line.

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## **Description**

### **BRIEF DESCRIPTION OF THE DRAWINGS**

- (1) FIG. 1 is a diagrammatic illustration of an exemplary disclosed additive manufacturing machine and a corresponding system that may be used to control the machine;
- (2) FIG. 2 is a schematic illustration of the control system of FIG. 1;
- (3) FIG. 3 is a flowchart representing an exemplary method that may be implemented by the control system of FIGS. 1 and 2;
- (4) FIGS. 4A, 5A and 6A are schematic illustrations of a region of a structure to be fabricated by the machine of FIG. 1;
- (5) FIGS. 4B, 5B and 6B are schematic illustrations of the regions of FIGS. 4A, 5A and 6A respectively including exemplary tool paths;
- (6) FIG. 7 is a flowchart representing an exemplary method that may be implemented by the control system of FIGS. 1 and 2;
- (7) FIG. 8 is a schematic illustration of a vector field through a region of a structure to be fabricated by the machine of FIG. 1;
- (8) FIG. 9 is a schematic illustration of exemplary tool paths to be used by the machine of FIG. 1 during fabrication of a structure;
- (9) FIG. 10 is a schematic illustration of a planar tool path projected onto a non-planar structure to be fabricated by the machine of FIG. 1;
- (10) FIGS. 11 and 12 are schematic representations of methods of assembling existing tool paths for use by the machine of FIG. 1;
- (11) FIG. 13 is a flowchart showing a method of assembling existing tool paths to create a desired shape of a structure to be fabricated by the machine of FIG. 1;
- (12) FIGS. 14-16 are schematic illustrations of arrangements for propagating layers of a structure to be fabricated by the machine of FIG. 1;
- (13) FIGS. 17-19 are schematic illustrations of a process for removing self-intersection from a propagated layer of a structure to be fabricated by the machine of FIG. 1;
- (14) FIGS. 20-23 are schematic illustrations showing a parameterization boundary used during propagation of layers of a structure to be fabricated by the machine of FIG. 1;

- (15) FIG. **24** is a flowchart representing an exemplary method that may be implemented by the control system of FIGS. **1** and **2**;
- (16) FIGS. **25a-f** are schematic illustrations of a virtual model of a layer of a structure at various stages of the method shown in FIG. **26**;
- (17) FIG. **26** is flowchart representing an exemplary method of determining a tool path for use by the machine of FIG. **1** during additive manufacturing a structure;
- (18) FIG. **27** is a schematic representation of regions of a layer of a structure to be fabricated by the machine of FIG. **1**;
- (19) FIG. **28** is a schematic illustration of an exemplary structure to be fabricated by the machine of FIG. **1**;
- (20) FIG. **29** is a schematic illustration of exemplary graph structures of a layer of a structure to be fabricated by the machine of FIG. **1**; and
- (21) FIG. **30** is a flowchart representing an exemplary method that may be implemented by the control system of FIGS. **1** and **2**.

#### DETAILED DESCRIPTION

- (22) FIG. **1** illustrates an exemplary control system (“system”) **10**, which may be used to design, plan, fabricate, and/or analyze a structure **12** having any desired shape, size, consist, and functionality. System **10** may include, among other things, an additive manufacturing machine (“machine”) **14** and at least one computing device **16** operatively connected to machine **14**. Machine **14** may be configured to create structure **12** under the guided control of computing device **16**, for example by way of an additive manufacturing process. Although additive manufacturing processes utilizing one or more continuous reinforcements (e.g., fibers—F) and one or more curable matrixes (M) will be described below as examples of how structure **12** may be created, it should be noted that other processes known in the art could alternatively be utilized for this purpose and benefit from the disclosed control systems and methods.
- (23) Machine **14** may be comprised of components that are controllable to create structure **12**, layer-by-layer and/or in free space (e.g., without the bracing of an underlying layer). These components may include, among other things, a support **18** and any number of heads **20** coupled to and/or powered by support **18**. In the disclosed embodiment of FIG. **1**, support **18** is a robotic arm capable of moving head **20** in multiple directions during fabrication of structure **12**. It should be noted that any other type of support (e.g., a gantry, an arm/gantry combination, etc.) capable of moving head **20** in the same or in a different manner could also be utilized, if desired.
- (24) Each head **20** (only one shown in FIG. **1**, for clarity) may be configured to discharge at least a matrix (e.g., a liquid resin, such as a zero volatile organic compound resin; a powdered metal; etc.) that is curable. Exemplary curable matrixes include a thermoset polymer, single- or multi-part epoxy resins, polyester resins, cationic epoxies, acrylated epoxies, urethanes, esters, a thermoplastic polymer, photopolymers, polyepoxides, thiols, alkenes, thiol-enes, and more. In one embodiment, the matrix inside each head **20** may be pressurized, for example by an external device (e.g., an extruder or another type of pump—not shown) that is fluidly connected to head **20** via a corresponding conduit (not shown). In another embodiment, however, the pressure may be generated completely inside of head **20** by a similar type of device. In yet other embodiments, the matrix may be gravity-fed through and/or mixed within head **20**. In some instances, the matrix inside head **20** may need to be kept cool and/or dark to inhibit premature curing; while in other instances, the matrix may need to be kept warm for the same reason. In either situation, head **20** may be specially configured (e.g., insulated, temperature-controlled, shielded, etc.) to provide for these needs.
- (25) In some embodiments, the matrix may be mixed with, contain, or otherwise at least partially wet or coat one or more reinforcements (e.g., powders, particles, chopped fibers, unchopped fibers, tows, braids, rovings, fabrics, knits, mats, socks, sheets, tubes, etc.) and, together with the reinforcements, make up at least a portion (e.g., a wall, a surface, internal fill, etc.) of structure **12**.

The reinforcements may be stored within (e.g., on one or more separate internal spools—not shown) or otherwise passed through head **20** (e.g., fed from one or more external spools). When multiple reinforcements are simultaneously used, the reinforcements may be of the same material and have the same sizing and cross-sectional dimension and shape, or a different material with different sizing and/or cross-sectional dimension and shape. The sizing may include, for example, treatment of the reinforcement with plasma, treatment with an acid (e.g., nitric acid), or otherwise surface-functionalized with an agent (e.g., a dialdehyde, an epoxy, a vinyl, and/or another functional group) to enhance adhesion of the matrix to the reinforcement. The reinforcements may include, for example, carbon fibers, vegetable fibers, wood fibers, mineral fibers, glass fibers, metallic wires, optical tubes, etc. It should be noted that the term “fiber” is meant to encompass both structural and non-structural (e.g., functional) types of continuous reinforcements that can be at least partially encased in the matrix discharging from head **20**.

(26) For the purposes of this disclosure, continuous reinforcements may be considered to have an aspect ratio ( $V$ ) defined as a length ( $L$ ) divided by a diameter ( $d$ ) (e.g.,  $V=L/d$ ) that is greater than 10, 100, 1000, 100,000, 1,000,000 or even larger. Discontinuous reinforcements may include reinforcements having an aspect ratio less than that of continuous reinforcements.

(27) The reinforcements may be exposed to (e.g., at least partially coated and/or internally wetted with) the matrix while the reinforcements are inside head **20**, while the reinforcements are passing into head **20**, and/or while the reinforcements are discharging from head **20**. The matrix, dry (e.g., unimpregnated) reinforcements, and/or reinforcements that are already exposed to the matrix (pre-impregnated reinforcements) may be transported into head **20** in any manner apparent to one skilled in the art. In some embodiments, discontinuous may be mixed with the matrix and/or additives before and/or after the matrix coats continuous reinforcements.

(28) Support **18** may move head **20** in a particular trajectory (e.g., a trajectory corresponding to an intended shape, size, and/or function of structure **12**) at the same time that the matrix-wetted reinforcement(s) discharge from head **20**, such that one or more continuous paths of matrix-wetted reinforcement(s) are formed along the trajectory. Each path may have any cross-sectional shape, diameter, and/or fiber-to-matrix ratio, and the reinforcements may be radially dispersed with the matrix, located at a general center thereof, or located only at a periphery.

(29) One or more cure enhancers (e.g., a light source, a radiation source, an ultrasonic emitter, a microwave generator, a magnetic field generator, a temperature regulator, a catalyst dispenser, a fan, etc.) **22** may be mounted proximate (e.g., within, on, and/or adjacent) head **20** and configured to affect (e.g., initiate, enhance, complete, or otherwise facilitate) curing of the matrix as it is discharged with the reinforcement(s) from head **20**. Each cure enhancer **22** may be independently and/or cooperatively controlled to selectively expose one or more portions of the discharging material to cure energy (e.g., electromagnetic radiation, vibrations, positive or negative thermal energy, a chemical catalyst, a pressurized medium, etc.). The energy may trigger a reaction to occur within the matrix, increase a rate of the reaction, sinter the matrix, pyrolyze the matrix, harden the matrix, stiffen the matrix, or otherwise cause the matrix to partially or fully cure as it discharges from head **20**. The amount of energy produced by cure enhancer **22** may be sufficient to at least partially cure the matrix before structure **12** axially grows more than a predetermined length away from head **20**. In one embodiment, structure **12** is cured sufficient to hold its shape before the axial growth length becomes equal to an external dimension of the matrix-coated reinforcement.

(30) In the embodiment of FIG. **1**, head **20** is modular. For example, head **20** may include a matrix reservoir **26** and an outlet **24** that is removably connected to matrix reservoir **26** (e.g., via one or more threaded fasteners, clasps, actuators, and/or other hardware—not shown). In this example, outlet **24** is a single-track outlet configured to discharge composite material having a generally circular, rectangular, triangular, and/or other polygonal cross-section. The configuration of head **20**, however, may allow outlet **24** to be swapped out for another outlet that discharges composite material having a different shape (e.g., a tubular cross-section, a ribbon or sheet cross-section, etc.)

and/or a different number of tracks.

(31) In some embodiments, cure enhancer(s) **22** may be mounted to or adjacent a lower portion of outlet **24**. With this configuration, cure enhancer(s) **22** may be located around a distal end in a configuration that best suits the shape, size, and/or type of material discharging from outlet **24**. In the disclosed embodiment, cure enhancer(s) **22** are mounted at an angle relative to a central axis of outlet **24**, such that energy from cure enhancer(s) **22** is directed toward the material discharging from outlet **24**. One or more optics **31** may be used in some applications, to selectively block, disperse, focus, and/or aim the energy from cure enhancers **22** toward a discharge location of outlet **24**. This may affect a cure rate of and/or cure location on the material discharging from outlet **24**. It is contemplated that optics **31** may be adjustable, if desired (e.g., manually adjustable via a set screw—not shown, or automatically adjustable via an actuator—not shown).

(32) The matrix and/or reinforcement may be discharged together from head **20** via any number of different modes of operation. In a first example mode of operation, the matrix and/or reinforcement are extruded (e.g., pushed under pressure and/or mechanical force) from head **20** as head **20** is moved by support **18** to create features of structure **12**. In a second example mode of operation, at least the reinforcement is pulled from head **20**, such that a tensile stress or drag force is created in the reinforcement during discharge. In this second mode of operation, the matrix may cling to the reinforcement and thereby also be pulled from head **20** along with the reinforcement, and/or the matrix may be discharged from head **20** under pressure along with the pulled reinforcement. In the second mode of operation, where the reinforcement is being pulled from head **20**, the resulting tension in the reinforcement may increase a strength of structure **12** (e.g., by aligning the reinforcements, inhibiting buckling, equally loading the reinforcements, etc.) after curing of the matrix, while also allowing for a greater length of unsupported structure **12** to have a straighter trajectory. That is, the tension in the reinforcement remaining after curing of the matrix may act against the force of gravity (e.g., directly and/or indirectly by creating moments that oppose gravity) to provide support for structure **12**.

(33) The reinforcement may be pulled from head **20** as a result of head **20** being moved and/or tilted by support **18** away from an anchor point (e.g., a print bed, an existing surface of structure **12**, a fixture, etc.) **32**. For example, at the start of structure formation, a length of matrix-impregnated reinforcement may be pulled and/or pushed from head **20**, deposited against anchor point **32**, and at least partially cured, such that the discharged material adheres (or is otherwise coupled) to the anchor point **32**. Thereafter, head **20** may be moved and/or tilted away from anchor point **32**, and the relative motion may cause the reinforcement to be pulled from head **20**. As will be explained in more detail below, the movement of reinforcement through head **20** may be selectively assisted via one or more internal feed mechanisms, if desired. However, the discharge rate of reinforcement from head **20** may primarily be the result of relative movement between head **20** and anchor point **32**, such that tension is created within the reinforcement. As discussed above, anchor point **32** could be moved away from head **20** instead of or in addition to head **20** being moved away from anchor point **32**.

(34) Any number of separate computing devices **16** may be used to design and/or control the placement of fibers within structure **12** and/or to analyze performance characteristics of structure **12** before, during, and/or after formation. Computing device **16** may include, among other things, a display **34**, one or more processors **36**, any number of input/output (“I/O”) devices **38**, any number of peripherals **40**, and one or more memories **42** for storing programs **44** and data **46**. Programs **44** may include, for example, any number of design, pathing, and/or printing apps **48** and an operating system **50**.

(35) Display **34** of computing device **16** may include a liquid crystal display (LCD), a light emitting diode (LED) screen, an organic light emitting diode (OLED) screen, and/or another known display device. Display **34** may be used for presentation of data under the control of processor **36**.

(36) Processor **36** may be a single or multi-core processor configured with virtual processing technologies and use logic to simultaneously execute and control any number of operations. Processor **36** may be configured to implement virtual machine or other known technologies to execute, control, run, manipulate, and store any number of software modules, applications, programs, etc. In addition, in some embodiments, processor **36** may include one or more specialized hardware, software, and/or firmware modules (not shown) specially configured with particular circuitry, instructions, algorithms, and/or data to perform functions of the disclosed methods. It is appreciated that other types of processor arrangements could be implemented that provide for the capabilities disclosed herein.

(37) Memory **42** can be a volatile or non-volatile, magnetic, semiconductor, tape, optical, removable, non-removable, or other type of storage device or tangible and/or non-transitory computer-readable medium that stores one or more executable programs **44**, such as design, analysis, pathing, monitoring and/or printing apps **48** and operating system **50**. Common forms of non-transitory media include, for example, a flash drive, a flexible disk, a hard disk, a solid-state drive, magnetic tape or other magnetic data storage medium, a CD-ROM or other optical data storage medium, any physical medium with patterns of holes, a RAM, a PROM, an EPROM, a FLASH-EPROM or other flash memory, NVRAM, a cache, a register or other memory chip or cartridge, and networked versions of the same.

(38) Memory **42** may store instructions that enable processor **36** to execute one or more applications, such as design, pathing, and/or fabrication apps **48**, operating system **50**, and any other type of application or software known to be available on computer systems. Alternatively or additionally, the instructions, application programs, etc. can be stored in an internal and/or external database (e.g., a cloud storage system—not shown) that is in direct communication with computing device **16**, such as one or more databases or memories accessible via one or more networks (not shown). Memory **42** can include one or more memory devices that store data and instructions used to perform one or more features of the disclosed embodiments. Memory **42** can also include any combination of one or more databases controlled by memory controller devices (e.g., servers, etc.) or software, such as document management systems, Microsoft SQL® databases, SharePoint® databases, Oracle® databases, Sybase® databases, or other relational databases.

(39) In some embodiments, computing device **16** is communicatively connected to one or more remote memory devices (e.g., remote databases—not shown) through a network (not shown). The remote memory devices can be configured to store information that computing device **16** can access and/or manage. By way of example, the remote memory devices could be document management systems, Microsoft SQL® database, SharePoint® databases, Oracle® databases, Sybase® databases, Cassandra®, HBase®, or other relational or non-relational databases or regular files. Systems and methods consistent with disclosed embodiments, however, are not limited to separate databases or even to the use of a database.

(40) Programs **44** may include one or more software or firmware modules causing processor **36** to perform one or more functions of the disclosed embodiments. Moreover, processor **36** can execute one or more programs located remotely from computing device **16**. For example, computing device **16** can access one or more remote programs that, when executed, perform functions related to disclosed embodiments. In some embodiments, programs **44** stored in memory **42** and executed by processor **36** can include one or more of design, pathing, fabrication, monitoring, and/or analysis apps **48** and operating system **50**. Apps **48** may cause processor **36** to perform one or more functions of the disclosed methods. Accordingly, embodiments may be embodied in hardware and/or in software (including firmware, resident software, micro-code, etc.) that runs on a processor, which may collectively be referred to as “circuitry,” “a module” or variants thereof.

(41) Operating system **50** may perform known operating system functions when executed by one or more processors, such as processor **36**. By way of example, operating system **50** may include Microsoft Windows®, Unix®, Linux®, OSX®, and IOS® operating systems, Android® operating



systems, or another type of operating system **50**. Accordingly, disclosed embodiments can operate and function with computer systems running any type of operating system **50**.

(42) Various methods and apparatus are described herein with reference to block diagrams or flowchart illustrations of computer-implemented methods, apparatus (systems and/or devices) and/or computer program products. It is understood that a block of the block diagrams and/or flowchart illustrations, and combinations of blocks in the block diagrams and/or flowchart illustrations, can be implemented by computer program instructions that are performed by one or more computer circuits. These computer program instructions may be provided to a processor circuit of a general purpose computer circuit, special purpose computer circuit, and/or other programmable data processing circuit to produce a machine, such that the instructions, which execute via the processor of the computer and/or other programmable data processing apparatus, transform and control transistors, values stored in memory locations, and other hardware components within such circuitry to implement the functions/acts specified in the block diagrams and/or flowchart block or blocks.

(43) It should also be noted that in some implementations, the functions/acts noted in the blocks may occur out of the order noted in the flowcharts. For example, two blocks shown in succession may in fact be executed substantially concurrently or the blocks may sometimes be executed in the reverse order, depending upon the functionality/acts involved. Moreover, the functionality of a given block of the flowcharts and/or block diagrams may be separated into multiple blocks and/or the functionality of two or more blocks of the flowcharts and/or block diagrams may be at least partially integrated. Finally, other blocks may be added/inserted between the blocks that are illustrated

(44) I/O devices **38** may include one or more interfaces for receiving signals, instructions, and/or other input from a user and/or machine **14**, and for providing signals, instructions and/or output to machine **14** that cause structure **12** to be printed. Computing device **16** can include, for example, components for interfacing with one or more input devices, such as one or more keyboards, mouse devices, and the like.

(45) Peripheral device(s) **40** may be standalone devices or devices that are embedded within or otherwise associated with machine **14** and used during fabrication of structure **12**. As shown in FIG. 2, peripherals **40** can embody input devices (e.g., one or more sensors, such as tension sensors, position sensors, pressure sensors, temperature sensors, proximity sensors, rotary encoders, scanners, and other sensors known in the art) **40A** and/or output devices (e.g., one or more actuators, such as a matrix supply, a fiber supply, a heater, a pump, cure enhancer **22**, a positioning motor, a cutter, a feed roller, a tensioner, a wetting mechanism, a compactor, etc.) **40B**. In some embodiments, peripherals **40** may, themselves, include one or more processors (e.g., a programmable logic control (PLC), a computer numeric controller (CNC), etc.), a memory, and/or a transceiver. When peripheral device(s) **40** are equipped with a dedicated processor and memory, the dedicated processor may be configured to execute instructions stored on the memory to receive commands from processor **36** associated with video, audio, other sensory data, control data, location data, etc., including capture commands, processing commands, motion commands, and/or transmission commands. The transceiver may include a wired or wireless communication device capable of transmitting data to or from one or more other components in system **10**. In some embodiments, the transceiver can receive data from processor **36**, including instructions for sensor and/or actuator activation and for the transmission of data via the transceiver. In response to the received instructions, the transceiver can packetize and transmit data between processor **36** and the other components.

(46) Design, pathing, fabrication, monitoring and/or analysis apps **48** may comprise computer executable instructions for performing all or part of the disclosed methods. Such methods may relate to generating, receiving, processing, analyzing, storing, and/or transmitting data in association with operation of machine **14** and corresponding

design/pathing/fabrication/monitoring/analysis of structure **12**. For example, apps **48** may comprise computer executable instructions for performing operations including one or more of: displaying a graphical user interface (GUI) on a display (e.g., display **34**) for receiving design/control instructions and information from a user; capturing sensory data associated with machine **14** (e.g., via peripherals **40A**); receiving instructions (e.g., via I/O devices **38** and/or the user interface) regarding specifications, desired characteristics, and/or desired performance of structure **12**; processing the control instructions; generating one or more possible designs of and/or plans for fabricating structure **12**; analyzing and/or optimizing the designs and/or plans; providing recommendations of one or more designs and/or plans; compiling and storing machine code including instructions for controlling machine **14** to fabricate a recommended and/or selected design via a recommended and/or selected plan; controlling machine **14** to fabricate a recommended and/or selected design via a recommended and/or selected plan; analyzing the fabrication; and/or providing feedback and adjustments to machine **14** for improving future fabrications. In exemplary arrangements, the design, pathing, fabrication, monitoring and/or analysis apps **48** may be executed, in whole or in part, by processor **36** of machine **14**. In other exemplary arrangements, the design, pathing, fabrication, monitoring and/or analysis apps **48** may be executed, in whole or in part, by a processor of a separate computing device remote from computing device **16**. In the latter arrangements, the separate computing device may store the compiled machine code for later transfer to machine **14**.

#### INDUSTRIAL APPLICABILITY

(47) The disclosed systems and methods may be used to manufacture composite structures having any desired cross-sectional shape, length, density, stiffness, strength, and/or other characteristic. The composite structures may be fabricated from any number of different reinforcements of the same or different types, diameters, shapes, configurations, and consists, and/or any number of different matrixes. Operation of system **10** will now be described in detail, with reference to the enclosed drawings.

(48) It has been found that a continuity of material discharge by machine **14** can affect a time required to fabricate structure **12** and/or a property of structure **12**. For example, the longer a path is continuously discharged, without having to slow or stop to sever the discharging material and restart discharging, the quicker structure **12** can be fabricated. Likewise, greater continuity may result in enhanced strength, stiffness, weight and/or other properties. FIGS. **3-6B** illustrate a method employed by computing device **16** that provides for fabrication of structure **12** with greater continuity. In addition, the methods described in relation to FIGS. **3-6B** allow for tool paths to be generated that place reinforcements in advantageous locations and orientations. This advantageous placement may enhance a desired property of structure **12**. For example, material may be located to accommodate a load that structure **12** is required to withstand.

(49) As can be seen in the flowchart of FIG. **3**, the first step may include receiving of boundary data (Step **300**), which in exemplary arrangements may include polygonal boundary data and/or spline data. The boundary data may be a two-dimensional outer boundary **100** of at least part of structure **12** at a particular height in a Z-direction. Outer boundary **100** may at least partially enclose a region **102** of structure **12**. Outer boundary **100** may be received as data by computing device **16** (e.g., by processor **36**) via a file (e.g., a CAD drawing), may be drawn by a user of computing device **16** (e.g., via I/O device(s) **38**), and/or may be selected from a variety of available boundaries. For example, the user may import, draw, and/or select the rectangularly shaped outer boundary **100** shown in dashed lines in FIG. **4A** or **5A** or the irregularly shaped outer boundary **100** of FIG. **6A**.

(50) As used herein, the term 'region' encompasses an area of structure **12** that is pathed separately from other regions **102** and optionally as a single tool path. A region **102** may define all or part of a layer of structure **12** and may be determined in a way that makes pathing simpler and/or improves a desired property of structure **12**. Regions **102** of structure **12** may be elongate. At least part of

different regions **102** of structure **12** may run transverse to each other and may meet each other and/or intersect. At least part of different regions **102** of structure **12** may run parallel to each other. (51) Computing device **16** may be configured to generate one or more tool paths **51** within region **102**. These tool path(s) **51** may guide machine **14** to discharge material at the boundary and thereby form a surface of structure **12** and/or inside of the boundary as infill that supports the surface. In some arrangements, tool path(s) **51** within a single layer of structure **12** may be arranged such that the surface and the infill are formed from a single continuous reinforcement. In some arrangements multiple segments of continuous reinforcements may be required and/or desired to separately form the surface and infill of a single layer.

(52) To generate tool path(s) **51**, computing device **16** may divide region **102** into two or more partitions **52** utilizing one or more partitioning lines **54** (Step **305**). Partitioning lines **54** may extend from the outer boundary **100** across at least part of region **102** and be straight or curved and 2D or 3D. In some arrangements, partitioning lines **54** may extend across the entire region **102** and cross the outer boundary **100** in two places (e.g., as in FIGS. **4A** and **5A**). In some arrangements, partitioning lines **54** may extend across only a part of region **102** (e.g., as in FIG. **6A**). Partitioning lines **54** may or may not form an enclosed area within region **102**. Multiple different types of partitioning lines **54** may be placed on region **102**, and each type may drive different operations.

(53) As shown in the exemplary arrangement of FIG. **4A**, a first type of partitioning lines **54** are placed in pairs. In this example, each pair of partitioning lines **54** may define an intersection lane **56** therebetween. Intersection lane **56** defines a location at which tool path **51** will cross region **102** and may define a location of an intersection of tool path(s) **51**. It should be noted that partitioning lines **54** making up intersection lane **56** need not be parallel. The placement of partitioning lines **54** relative to each other and relative to the outer boundary **100** of region **102** that they cross may affect a location, orientation, and/or curvature of the tool path(s) **51** that will intersect within each lane **56**. It is noted that the pairs of partitioning lines **54** defining intersection lanes **56** shown in FIG. **4A** may also be utilized in the irregular boundary and seam arrangement of FIG. **6A** (discussed below).

(54) As shown in the exemplary embodiment of FIG. **5A**, a second type of partitioning line **54** is placed as an individual line independent of other partitioning lines **54** within the same region. In this example, each partitioning line **54** may itself define a location at which tool path(s) **51** cross region **102** and may define a location of an intersection of the tool path(s) **51**. As in the example of FIG. **4A**, the placement of partitioning lines **54** relative to each other and relative to the outer boundary **100** of the polygon region that they cross may affect a location, orientation, and/or curvature of the infill tool paths **51** within partitions **52**. It is contemplated that computing device **16** may cause partitioning lines **54** to be shown on display **34** in different ways (e.g., in different colors, different line styles or weights, etc.) to signify characteristics of the particular type of partitioning line **54** used.

(55) Many different strategies may be utilized to place partitioning lines **54**. In one example, individual or paired partitioning lines **54** are manually placed across the boundary of region **102** (e.g., via I/O device(s) **38**). Partitioning lines **54** may be placed to achieve a desired load that structure **12** is required to withstand, a desired appearance, a desired infill shape or density, a desired strength, a desired weight distribution, a desired intersection with or avoidance of a feature, or another desired property. In another example, a template including a plurality of partitioning lines **54** may be selected from among a variety of different templates. The template may be positioned, oriented and scaled over region **102** based on the desired property. In another example, default partitioning lines **54** may be automatically placed in a default manner. Thereafter, adjustments may be made to the position, orientation, scale, etc. (e.g., based on the desired property). In yet another example, partitioning lines **54** may be placed based on a shape of region **102**, a size of region **102**, features of region **102**, and/or results of an analysis. The placement may be undertaken manually by a user or may be undertaken by processor **36** based on data relating to

the desired property.

(56) The analysis may include modeling a behavior of structure **12** under a given condition. This may allow an estimate of the desired property to be determined. In some arrangements, modeling a behavior of structure **12** may include generating a property (e.g., force, displacement, heat transfer, conductivity, radiation, etc.) vector field through region **102** and/or undertaking a finite element analysis (FEA) of region **102**. In these examples, partitioning lines **54** and/or the template may be positioned, oriented and/or scaled based on the vector field and/or a distribution map from the FEA. For example, partitioning lines **54** may be positioned, oriented and/or scaled such that sections of tool path(s) **51** are substantially aligned with a direction of one or more property vectors. Such alignment may be parallel to the direction of the one or more property vectors or aligned with the direction of the one or more property vectors within a predefined angle. In a further example, partitioning lines **54** may be positioned, oriented and/or scaled such that the subsequently fabricated structure **12** is able to withstand a required load or force.

(57) After partitioning lines **54** have been extended across region **102**, computing device **16** may generate one or more tool path(s) **51** (step **310**). As discussed above, tool path(s) **51** may navigate some or all of the outer boundary **100** and/or cross through region **102** based on the placement of the partitioning lines **54**. In one arrangement, an exemplary tool path **51** circumnavigates region **102** and crosses region **102** at lane(s) **56** and/or location(s) where partitioning line(s) **54** have been placed.

(58) Referring to FIGS. **4A** and **4B**, computing device **16** may apply a walking strategy to generate a tool path **51** that passes around region **102** and through partitions **52** and/or intersection lanes **56** (Step **310**). This may include, for example, determining a starting point (SP) for a given tool path **51**. The starting point may be selected by a user, generated by computing device **16** based on a desired property, and/or generated by computing device **16** and adjusted by a user. The starting point may lie on the outer boundary **100** of region **102**. It should be noted that computing device **16** may be configured to inhibit starting points of adjacent layers of structure **12** from aligning with each other, as this alignment could result in undesired properties within structure **12**.

(59) From the starting point, computing device **16** may extend tool path **51** along the outer boundary **100**. In one arrangement, a direction of tool path **51** away from the starting point may alternate between layers or stacks of layers (e.g., extending to the right relative to the perspective of FIGS. **4B** and **5B**, and then to the left in a subsequent layer or subsequent stack of layers). Tool path **51** may be shown by computing device **16** in a manner (e.g., color, line style, etc.) different from the boundary lines of the polygon and partitioning lines **54**.

(60) Tool path **51** extends away from the starting point and, based on the positioning of a partitioning line(s) **54**, tool path **51** may be steered away from the boundary line and across region **102**. This may occur at the point where tool path **51** encounters a partitioning line **54** or at a predetermined point before tool path **51** encounters a partitioning line **54**. It will be appreciated that the predetermined point may be determined based on a minimum or desired bend radius of tool path **51**. For the remainder of the description of FIGS. **4A-6B**, the predetermined point is not referred to, although it will be appreciated that it might be applied at any junction where there is a deviation of tool path **51**. As shown in FIG. **4B**, when the encountered partitioning line **54** is of the paired type, tool path **51** may be extended from the junction of the original boundary line and partitioning line **54** diagonally across intersection lane **56** to a catcorner junction. As shown in FIG. **5B**, when the encountered partitioning line **54** is of the independent type, tool path **51** may be extended across region **102** at least partially along partitioning line **54**. In exemplary arrangements, tool path **51** may follow the entire partitioning line **54** across region **102** (accounting for minimum bend radius). In exemplary arrangements, tool path **51** may extend from the junction of the original boundary line and partitioning line **54** diagonally across only the partitioning line **54**. Referring again to FIG. **4B**, when tool path **51** encounters the outer boundary **100** at the catcorner junction, computing device **16** may deviate tool path **51**, such that it extends away from the catcorner

junction along the newly encountered boundary line. This deviation may be in a direction to continue circumnavigation of region **102**. Specifically, relative to the current direction of propagation of tool path **51**, deviation away from the outer boundary **100** across region **102** may be in a first direction left or right. After the deviation left or right, the current direction of propagation of tool path **51** has changed as it progresses across the region. Deviation from region **102** to align with the newly encountered outer boundary **100** may be in the opposite direction (left or right) relative to the current direction of tool path **51**. That is, if the deviation away from the outer boundary **100** is left (relative to the direction of tool path **51** as it follows the outer boundary **100**) then the deviation of tool path **51** as it aligns with the newly encountered outer boundary **100** is right (relative to the direction of tool path **51** as it crosses region **102**), and vice-versa. This may continue around region **102** until tool path **51** returns to the starting point.

(61) Computing device **16** may then identify any intersections where the tool path **51** crosses over itself or over another tool path **51** within the same layer (Step **315**). This will occur, for example, within intersection lanes **56** and along the independent partitioning lines **54**. In order to maintain a height or z-dimension at each intersection that is substantially the same (e.g., within engineering tolerances) as a height or z-dimension at the outer boundary **100** over a plurality of layers, the intersection location of tool path **51** may be selectively split and/or shifted between differing layers (Step **320**).

(62) In arrangements, legs of an intersection in a first layer may be laterally shifted relative to corresponding legs in a second layer of structure **12**. For example, when completing tool path **51** of a first layer, one or both of a first leg and a subsequent second leg of tool path **51** may be laterally shifted an offset distance to one side (e.g., to the left) of a desired intersection location (shown in solid lines within enlarged portion of FIG. **4B**). Likewise, when completing a subsequent layer, one or both of the first and second legs of tool path **51** may be shifted to an opposing side (e.g., to the right) of the desired intersection location (shown in dashed lines within enlarged portion of FIG. **4B**). The shifting directions and/or offset amounts may be varied between the first and second legs and/or between alternating layers of structure **12**, such that for every set number of (e.g., for every two) layers of structure produced by machine **14**, the intersection location of tool path **51** increments in the height or z-dimension direction by a thickness of the set number of layers (e.g., by two diameters of the corresponding reinforcement). In some arrangements, the aggregated height or z-dimension at the intersection may be substantially the same as the aggregated height or z-dimension over of the set number of layers.

(63) It should be noted that, in some instances, machine **14** may not be capable of depositing reinforcement(s) along a particular tool path **51** generated via Steps **300-320**. For example, tool path **51** may have junctions and/or intersections that are too sharp or have too small of a radius for support **18** to accurately move print head **20** along. In another example, the reinforcement being used may have a minimum bend radius or other limitation that prevents its placement along tool path **51**. In yet another example, the matrix being used may have a minimum bend radius required for sufficient tack of the reinforcement. In any of these examples and others, tool path **51** may be manipulated by computing device **16** to conform to any associated limitations (Step **325**). As shown in FIGS. **4B** and **5B**, this may include computing device **16** smoothing out junctions and/or intersections, tightening or loosening of radiuses, adding or subtracting of tool path passes along boundary lines, shifting of tool path segments, etc. based on one or more pre-programmed limitations of machine **14** and/or the materials to be discharged by machine **14** along tool paths **51**. In exemplary arrangements, constraints related to tool path(s) **51** along which machine **14** is capable of depositing reinforcement(s) may be known and tool path(s) **51** may be generated such that those constraints are met.

(64) In exemplary instances, a user may be unsatisfied with the particular tool path **51** generated via steps **300-320**. For example, a user may desire a different intersection location, a different curvature, a thickness of walls at the outer boundary **100** of region **102** within each layer,

avoidance of a region feature, closer approximation of region **102** shape, etc. In any of these examples, computing device **16** may be configured to show tool path(s) **51** on display **34** and receive input from the user regarding desired changes (Step **330**).

(65) After manipulation of previously generated tool path(s) **51** (if any) either by a user or computing device **16**, tool path(s) **51** may be output for subsequent use by machine **14** in fabricating structure **12** (Step **335**). For example, machine code may be compiled, stored within a memory device, transmitted to machine **14**, or otherwise output in a manner that is suitable for controlling machine **14** to deposit material along tool path **51**.

(66) It is contemplated that the exact same tool path **51** may be utilized for every layer of structure **12**, that tool paths **51** for each layer of structure **12** may all be different, that some tool paths **51** for layers of structure **12** may be identical and others different, and/or that tool paths **51** for layers of structure **12** may scale, merge, and/or split at different layers. For example, in one application, it may be desirable for structure **12** to taper along a height or z-direction. In this example, after advancing the taper to a certain layer, a region that previously was large enough to accommodate a particular toolpath **51** could become too small. In this situation, one or more partitions **52** and/or intersection lanes **56** may need to merge or absorb into an adjacent partition **52** and/or intersection lane **56**. Similarly, a reverse taper of structure **12** may create partitions **52** and/or intersection lanes **56** that become too large at a particular layer for a given tool path **51** to fill adequately. In this example, the partition(s) **52** and/or intersection lane(s) **56** may need to be split apart into smaller partition(s) **52** and/or intersection lane(s) **56**. In one application, it may be possible for growing or shrinking layers to have tool paths **51** that overlap in undesirable ways (e.g., that extend too far unsupported).

(67) FIGS. **6A** and **6B** illustrate an exemplary irregular outer boundary **100**. In this instance, following Steps **300-335** may not generate a tool path **51** that is acceptable to the user. It is contemplated that prior to executing Step **305** (e.g., prior to extending partitioning lines **54**), region **102** may be selectively adjusted. For example, a seam **58** may be created inside of region **102**, and the partitioning lines **54** may thereafter be extended from one edge of the outer boundary **100** to the newly created seam **58**. It is contemplated that seam **58** may have a zero or non-zero volume, as desired, and any size or shape selected or created by the user. In one example, seam **58** is open to an exterior of region **102** and closed to an interior of region **102**, such that a subsequently generated tool path **51** will follow along an internal side of seam **58** as it would a normal section of the outer boundary **100** of region **102**. In another example (not shown), seam **58** may be completely enclosed and placed entirely within the outer boundary **100** of region **102** to form an island, to which partitioning lines **54** may be extended.

(68) As indicated above, in regard to Step **305**, it is contemplated that partitioning lines **54** may be placed by computing device **16** across region **102** based on analysis. FIGS. **7** and **8** provide and illustrate an exemplary method for doing this. As shown in FIGS. **7** and **8**, a first step may include modeling behavior of structure **12** under expected loading conditions during intended use (Step **700**). Computing device **16** may then responsively determine, for the desired property, one or more vector field(s) passing through region **102** at the given layer of structure **12** (Step **705**). In one example, computing device **16** may generate a force-based vector field (e.g., a constant linear field, a cylindrical field, a radial field, and/or a complex field of multiple types) through the polygon using the load conditions received at Step **700**. This analysis may be simple and require little processing time and power, as high-accuracy in these initial results may not be critical.

(69) In some embodiments, generation of the vector field may be customizable. For example, the user may have an option of choosing different relationships governing how the vector field is to be generated. For example, the user may be able to select a decay rate of forces (or other parameters) within the vector field (e.g., based on a distance from a location at which the loading condition(s) are applied) or stipulate that the vector profile should be cylindrical, linear, and/or radial. It is also contemplated that the vector field(s) may be generated via user input (i.e., the user may be able to

draw or otherwise place the vector field). Alternatively, after generation of the vector field(s), the user may be able to manually adjust (e.g., push, pull, rotate, stretch, etc.) the field(s). In yet other embodiments, the vector field(s) may simply be imported from a file. The vector field is represented in FIG. 8 by arrows 79.

(70) As discussed above, computing device 16 may be configured to then generate a virtual layout (e.g., trajectories) of partitioning lines 54 relative to region 102 based on the vector field, or other analysis. In the example of FIG. 8, partitioning lines 54 are substantially aligned with arrows 79 representing the direction of the vector field. As discussed above, partitioning lines 54 may divide region 102 into partitions 52 and be used to generate tool path(s) 51 via the method of FIG. 3 discussed above (Step 710). Computing device 16 may generate the layout by generally aligning trajectories of partitioning lines 54 with the field vectors and spacing partitioning lines 54 based on a magnitude of the vectors, for example having a smaller spacing as magnitude increases and vice-versa. Partitioning lines 54 may be positioned, oriented and/or scaled so as to generate sections of tool path 51 crossing region 102 that are generally aligned with the vectors and spaced based on a magnitude of the vectors. It should be noted that the vector field(s) may be as coarse or refined as desired. However, when a coarser vector field is utilized, some interpolation between vectors in the field may be required when aligning and spacing partitioning lines 54.

(71) After completion of Step 710, computing device 16 may generate a mesh (e.g., a 2D mesh) across a surface of region 102 and render the results on display 34 (Step 715). This may provide an opportunity for the user to make adjustments to the mesh prior to performance of an analysis routine (e.g., FEA—Step 720). It should be noted that the analysis of Step 720 may be conducted based on the layout of reinforcements crossing region 102 and navigating along the outer boundary 100 of region 102 (i.e., as defined by tool path(s) 51) and the associated changes to material properties afforded by the reinforcements placed along the tool paths. That is, the analysis of Step 720 may be performed on region 102, accounting for placement of the continuous reinforcements along the tool path(s) 51. Comparison of the analysis results with one or more threshold requirements may be used to determine if tool path(s) 51 are acceptable (Step 725). When the layout of tool path(s) 51 is acceptable (Step 725:Y), computing device 16 may output the tool path(s) 51 in the manner discussed above.

(72) If at Step 725, it is determined that tool path(s) 51 do not produce desired results (Step 725:N), control may advance to Step 740 instead of Step 730. At Step 740, computing device may be configured to selectively adjust one or more of partitioning lines 54. In one example, at least one end of at least one partitioning line 54 may be selectively stepped along a direction of the associated outer boundary 100. In some instances, both ends of the partitioning line 54 may be stepped (e.g., by the same or different amounts and in the same or different directions), such that the partitioning line is rotated and/or shifted. When both ends of a partitioning line 54 are stepped, they may be stepped independently or simultaneously. After each step enacted at Step 740, control may return to Step 720, where the analysis is performed again. This may continue until performance of tool path(s) 51 is determined to be acceptable at Step 725. In some arrangements, the shape and/or type of partitioning line 54 may be altered at Step 740.

(73) In some applications, it may be desirable to generate multiple separate tool paths 51 for fabricating a region of a layer. These tool paths 51 may be generated for a region via partitioning as discussed above and/or created via other means (e.g., via a completely manual process, via a different automated or semi-automated algorithm, etc.). In the example of FIG. 9, a first or inner tool path 51A generated via the process of FIG. 3 is used to fabricate infill for structure 12, while a second or outer tool path 51B is used to generate an outer wall, surface, or skin. The second tool path 51B circumnavigates at least part of the outer boundary 100 of region 102 and, in one exemplary arrangement, circumnavigates the entire outer boundary 100. In one example, the second tool path 51B may be generated by simply tracing around the outer boundary 100 of a given region. Tool paths 51A and 51B may be selectively nested within a single layer of structure 12 to

include both the infill and the skin. Tool paths **51A** and **51B** may lie close enough to each other, such that they bond during discharge. Alternatively, tool paths **51A** and **51B** may be spaced apart from each other and remain separate within the layer. In this latter arrangement, an additional tool path may be deposited between tool paths **51A** and **51B** to bond tool paths **51A** and **51B** to each other. Alternatively, an additional tool path may be deposited in an adjacent layer to bond tool paths **51A** and **51B** to each other. For example, the additional tool path in the adjacent layer may cross over portions of both tool paths **51A** and **51B**, thereby bonding them together.

(74) There are different options for usage of tool paths **51A** and **51B**. For example, one of each of tool paths **51A** and **51B** may be used within separate layers of structure **12**. Alternatively or additionally, usage of tool paths **51A** and **51B** may be alternated between layers or between stacks of layers. In yet another example, tool path **51A** may be used alone to fabricate one or more adjacent layers of structure **12**, followed by tool path **51B** being used alone to fabricate one or more adjacent layers of structure **12**, and/or followed by both tool paths **51A** and **51B** being used together to fabricate one or more adjacent layers. Sequences of any number of different tool paths **51** may also be utilized. Many different combinations are possible.

(75) In some instances, it may be desirable to start with multiple existing tool paths **51** and assemble them in different ways to make a desired overall shape of structure **12**. An exemplary process is depicted in FIGS. **11** and **12** and outlined in the flowchart of FIG. **13**. FIG. **11** shows a plurality of different regional tool paths **1400A**, **1400B** and **1400C**. As used herein, the term ‘regional tool path’ encompasses a tool path that may be used by additive manufacturing machine **14** to manufacture a specific region of structure **12**. The regional tool paths **1400A**, **1400B**, **1400C** may include tool paths generated using any of the methods disclosed herein.

(76) As shown in the flowchart of FIG. **13**, the first step(s) to assemble a desired overall shape of structure **12** may include receiving any number of existing regional tool paths **1400A**, **1400B**, **1400C** (Step **1600**) to be used along with a desired number of instances where the regional tool paths **1400A**, **1400B**, **1400C** should be duplicated to produce one or more desired heights of structure **12** (Step **1605**). In the example of FIG. **11**, three different regional tool paths **1400A**, **1400B**, **1400C** have been received by computing device **16** and arranged into three different nodes Node **1**, Node **2**, and Node **3**. For example, Node **1** includes six layers of tool path **1400A**, Node **2** includes 12 layers of tool path **1400B**, and Node **3** includes five layers of tool path **1400C**. Note that the nodes do not need to have the same numbers of layers and/or the same heights (**H**).

(77) When the nodes are built up from regional tool paths **1400A**, **1400B**, **1400C** by the user and provided to computing device **16**, a placement of each node of tool paths within a virtual environment may also be determined by the user. The placement of the nodes may be provided to computing device **16** in the form of placement data, including one or more of a relative lateral position, a relative height position and a relative orientation. The placement data may therefore include relative coordinates and orientations. The user may input the placement data in different ways, such as by manual entering of the coordinates and orientations, drag-and-drop of node representations, and/or sketching of the nodes within the virtual environment.

(78) After receiving the composition and placement data for the nodes from the user, computing device **16** may generate corresponding labels and place the labels within an organizational tree shown on display **34** (Step **1610**). In some embodiments, an order of the labels within the tree may correspond with a physical stacking order of the corresponding nodes in a particular direction (e.g., within the height direction) within the virtual environment. For example, Node **3** (including tool paths **1400C**) is listed uppermost in the tree of FIG. **12**, thereby designating Node **3** as being located at a highest location in the stacking direction. Similarly, Node **2** (including tool paths **1400B**) is located lowest and Node **1** (including tool paths **1400A**) is located between Nodes **2** and **3**. In these embodiments, the user may be able to adjust the order of the labels within the tree (e.g., via drag-n-drop commands of the user received by computing device **16**—Step **1615**) to thereby change the physical order of the nodes and corresponding stacks of regional tool paths **1400A**,



**1400B, 1400C** in the height direction within the virtual environment. It is also contemplated that the user may be able to adjust a relative location and/or orientation of the nodes in dimensions other than the height direction, if desired.

(79) Once regional tool paths **1400A, 1400B, 1400C** have been received by computing device **16** and arranged into nodes within the organizational tree of FIG. **12**, computing device **16** may receive one or more operations from the user that relate regional tool paths **1400A, 1400B, 1400C** and/or nodes to each other (Step **1620**). The operations may be provided to computing device **16** in the form of operation data that is indicative of an operation to be performed in relation to at least one node, and in some cases on two or more nodes. For example, a ‘reverse’ operation may be undertaken on a single node that reverses the order of the regional tool paths in a given node, and a ‘translate’ operation may be undertaken on a single node that forms a translated copy of the node (i.e., a copy of the node is positioned elsewhere in the virtual environment). Also, an interleave operation (Operation **1** from FIG. **11**) or a stack operation (Operation **2** from FIG. **11**) may be required for two or more nodes. The operation data may include one or more operations for connecting the nodes together.

(80) The interleave operation is shown as relating Node **1** (including multiple copies of tool path **1400A**) to Node **2** (including multiple copies of tool path **1400B**). In this operation, data is created that defines one or more layers of structure **12** assembled from a copy of a layer from Node **1** and a copy of a corresponding layer of Node **2**. That is, the data representing the individual regional tool paths **1400A, 1400B** from a layer at a common height is combined into data representing a single tool path including the regional tool paths **1400A, 1400B** from Nodes **1** and **2** at the same height.

(81) In one illustrated example **1402** shown in the upper right corner of FIG. **11**, sequentially overlapping layers of structure **12** include a regional tool path copy from each of Node **1** and Node **2** (i.e., one planar or non-planar layer compiled from two different regional tool paths **1400A, 1400B**). In this example, layers of structure **12** including copies of tool path **1400A** and tool path **1400B** may extend only to a height of the shortest Node (in this case, Node **1**) and the regional tool paths **1400B** of the taller Node (in this case, Node **2**) may extend beyond the final layer of Node **1**. In another example **1404** shown below example **1402** in FIG. **11**, only particular layers of structure **12** include a regional tool path copy from each of Node **1** and Node **2**, while other layers include only a regional tool path copy from Node **1**. In this example, sequential tool path copies from the shortest Node (in this case, tool paths **1400A**) may be spaced apart from each other in a height or z-dimension (e.g., by two or more layers).

(82) The stack operation is shown in FIG. **11** as relating the compilation of Node **1** and Node **2** to Node **3**. In this operation, the layers of Node **3** are simply stacked at some height above the compilation of Node **1** and Node **2**. In some embodiments, Node **3** is positioned such that the lowest layer of Node **3** is stacked directly on top of a highest layer of the compilation. In other embodiments, Node **3** is positioned such that the lowest layer of Node **3** is spaced apart from the highest layer of the compilation (e.g., such that a space exists therebetween).

(83) The organizational tree includes the operations to be performed, the node(s) on which the operations are to be undertaken, an order in which nodes are to be positioned, and an order in which the operations are to be undertaken. Completion of all the user-defined operations in the tree may be considered “resolving” the regional tool paths (Step **1625**) into a fewer number of (e.g., one or more) resolved tool path(s) **1408**. Resolving the tool paths may serve to bond the regional tool paths **1400A, 1400B, 1400C** together.

(84) It should be noted that, in some instances, the resolved tool path(s) **1408** may not yet be ready for output to and consumption by machine **14**. For instance, gaps and/or overhangs between and/or under the tool paths **1400A, 1400B, 1400C** may exist that would cause machine **14** to fail in fabrication of structure **12** using the resolved tool path(s) **1408**. Accordingly, computing device **16** may determine a suitability of resolved tool path(s) **1408** for use with machine **14** based on one or more predefined criteria (Step **1630**). For example, computing device **16** may check for gaps and/or

overhangs that exceed allowable thresholds, although other criteria may be used. The predefined criteria may relate to operational constraints of machine **14** and/or a property of structure **12** during and/or after manufacture.

(85) Computing device **16** may adapt resolved tool path(s) **1408** based on the determined suitability (Step **1635**). In some exemplary arrangements, adapting the resolved tool path(s) may include amending placement and/or operation data. In some exemplary arrangements, adapting resolved tool path(s) **1408** includes creating a support tool path. In cases where a gap is identified, the support tool path may connect the two or more regional tool paths between which the gap exists. In cases where an identified overhang exceeds an overhang threshold value, the support tool path (e.g., support path **1410**) may be formed to support or brace the overhanging part of the resolved tool path(s) **1408**.

(86) Before or after completion of Steps **1630** and **1635**, computing device **16** may be configured to consider an aspect ratio of the structure **12** to be fabricated using the resolved tool path(s) **1408** (Step **1640**). In embodiments, this may be undertaken as part of determining suitability of resolved path(s) **1408**. For example, during fabrication of structure **12**, machine **14** may deposit material along the resolved tool path(s) **1408** that creates separate free-standing features of structure **12**. In some applications, the free-standing features may be tethered to each other at every layer. In other applications, however, the free-standing structures may be connected only after multiple layers have been deposited. In these latter applications, the features could become unstable at a given height prior to connection if the aspect ratio of the features is too high. Accordingly, computing device **16** may consider the aspect ratio of features of structure **12** during fabrication based on the resolved tool paths) **1408**. Based on the determined aspect ratio (e.g., if the determined aspect ratio is above an aspect ratio threshold value), computing device **16** may selectively generate support tool path(s) or extend (or otherwise amend) existing parts of resolved tool path(s) **1408** to brace or tether the high aspect ratio features. For example, the generated support tool path(s) and/or the extended existing parts of resolved tool path(s) **1408** may interconnect high aspect ratio features with other parts of resolved tool path(s) **1408** at particular locations within particular layers (Step **1645**). It is contemplated that this interconnection may be implemented by computing device **16**, implemented under the guidance of a user, or a combination of these methods.

(87) For example, the user may be able to instruct that high-aspect ratio features of the resolved tool path(s) **1408** are interconnected within every layer of structure **12**, that only particular high aspect ratio features of the resolved tool path(s) **1408** are interconnected within every layer, and/or that only particular high aspect ratio features of the resolved tool path(s) **1408** are interconnected within only particular layers. In addition, the user may allow or disallow automatic interconnection in addition to or instead of manual interconnection. In some instances, the user may be able to instruct where and/or how interconnection is to occur, while in other instances, computing device **16** may make these decisions based on efficiency, strength, material usage, and/or other considerations. Control may proceed from Step **1645** to Step **1625**, at which any interconnections are resolved with the existing resolved tool path(s) **1408**.

(88) It is noted that in other embodiments, determining the suitability of a tool path may be embodied in an operation that a user may place in the organizational tree of FIG. **12** and that may be resolved along with the other operations. For example, there may exist a “support overhang” operation, a “fill gap” operation and/or an “aspect ratio” operation.

(89) After completion of Steps **1640**:Y, computing device **16** may compile and output machine code including instructions for controlling machine **14** based on resolved tool path(s) **1408**. The compiled instructions may be stored and/or used by computing device **16** to control machine **14** to manufacture the structure based on resolved tool path(s) **1408**. Machine **14** may use these tool paths **51** to regulate motion of support **18** and head **20**.

(90) FIGS. **14**, **15** and **16** illustrate a method of determining tool paths for use in additively manufacturing structure **12**. In one example, a propagation surface may be established via data

received (Step **2400**) by computing device **16**. The propagation surface may form a base onto which material may be deposited in layers to build up structure **12** in a propagation direction. The propagation surface may correspond with an exposed surface **60** of a build platform or other anchor point **32** and/or an existing surface of structure **12**.

(91) One or more propagation boundaries **62**, **64**, **66** may also be determined (Step **2402**). Each of the propagation boundaries **62**, **64**, **66** may be identified as a mesh associated with a model of structure **12** in a virtual environment. The mesh may be determined by computing device **16** using any of the methods described herein and/or known in the art, or may be determined by a user and received as data by computing device **16**. One or more of the propagation boundaries **62**, **64**, and/or **66**, together with the propagation surface, may define an enclosed volume.

(92) It will be appreciated that each propagation boundary may have any shape (e.g., 2D or 3D) and be formed by placement on, adjacent and/or around propagation surface **60** and/or via projection toward propagation surface **60**. For example, propagation boundary **62** may be a five-sided irregular box (e.g., including surfaces **62A** and **62B**) that is placed on propagation surface **60** to enclose a volume. Alternatively, propagation boundary **62** may simply be a three-dimensional surface or outline (e.g., **62B**) that is projected toward propagation surface **60** (e.g., in a direction normal to propagation surface **60** or surface **62B** at a particular point in space), thereby defining the same or another enclosed volume defined by the irregular box. Other methods for defining a propagation boundary may also be utilized.

(93) A portion of propagation surface **60** that is contained by the boundary **62** (e.g., within the side walls of surface **62A**) may define an area on which material is to be deposited when manufacturing structure **12**. As discussed below, each structure layer of structure **12** may be determined such that it propagates through the enclosed volume in a propagation direction **70** (referring to FIG. **15**). Propagation direction **70** may be defined by propagation surface **60** and, as in the case of FIG. **14**, may be substantially normal to the surface. In the simple case of planar propagation surface **60**, offsetting each structure layer may be a simple process. But as discussed below, in other cases, some correction may be required.

(94) The first layer of material (structure layer) to be deposited on propagation surface **60** may be considered a base layer **68**. Base layer **68** is offset from propagation surface **60** in propagation direction **70** by a distance equivalent to a thickness or height of a material to be deposited. Base layer **68** is contained within boundary **62**. That is, boundary **62** defines a lateral extent of base layer **68**. Base layer **68** may be 2-dimensional or 3-dimensional (i.e., non-planar). In instances when base layer **68** is 3-dimensional, base layer **68** may be mapped into 2-dimensions prior to pathing. Base layer **68** may be mapped into 2-dimensions using a surface or mesh parameterization algorithm, for example UV parameterization. UV parameterization maps a 3-dimensional surface to 2-dimensional space by trading off angular and spatial relationships of features of the 3-dimensional contour after it is mapped to 2-dimensions. That is, if an angle between features on a 3-dimensional surface are preserved when mapping the surface to 2-dimensional space, then a spatial relationship between those features will be distorted, and vice-versa. This can be seen in the example of the globe parameterized to a 2-dimensional world map, and in which certain countries appear larger than they actually are because angular relationships have been preserved at the expense of spatial relationships. In exemplary arrangements, surface parameterization may be undertaken so as to preserve a spatial relationship of features of base layer **68** (or other structure layers **72**), as opposed to preserving angular relationships. This means that tool paths generated in 2-dimensional space as adjacent to one another, remain adjacent to one another when they are mapped back to the 3-dimensional surface. The mapping may include discretizing the 3-dimensional base layer into points in a local, 2-dimensional coordinate system.

(95) A tool path may then be generated (Step **2408**) for the base layer **68** and may be considered a base tool path. The base tool path is a virtual trajectory along which machine **14** should deposit material during fabrication of the base layer **68** of structure **12**. The base tool path may be

generated using any method described herein, in whole or in part, or using any method known in the art.

(96) Additional overlapping structure layers may be generated by offsetting points from propagation surface **60** or base layer **68**, or by offsetting from any previous structure layer of structure **12** in propagation direction **70**. Offsetting layers may result in self-intersection in the offset layer (e.g., when the base layer includes concave features). This self-intersection may be corrected, as shown in FIGS. **17-19**. Referring to FIG. **17**, a layer of structure **12** is to be determined by offsetting a surface **1700** in a direction normal to that surface. The direction normal to surface **1700** is shown by arrows **1702**. FIG. **18** shows the resulting surface **1704**. Because surface **1700** includes a concave feature, when it is offset then one side **1700a** of the concave feature crosses an opposite side **1700b** of the concave feature. This results in self-intersection of resulting surface **1704**, which can be seen to self-intersect at point **1706** and the looped part **1708** of surface **1704** that is below the self-intersection point **1706**. The self-intersection, when seen across an entire surface, will, in most cases, be a seam across that surface. The self-intersection is identified by computing device **16** and the part **1708** of surface **1704** that is below self-intersection point **1706** is removed. The final offset surface **1710** is shown in FIG. **19**. Surface **1710** is parameterized and used to generate tool paths.

(97) Computing device **16** may determine a plurality of additional structure layers **72** (Step **2410**). Structure layers **72** may each be offset from propagation surface **60** or from a previous layer. For example, a first structure layer **72A** may be offset from base layer **68** and a second structure layer **72B** may be offset from first structure layer **72A**. Each structure layer **72** may be offset from the previous layer by an amount equal to a desired thickness of the structure layer **72** (e.g., a thickness of material to be deposited within the layer). It will be appreciated that one or more structure layers **72** may have substantially the same thickness. It will also be understood that thickness may vary between structure layers **72**. Computing device **16** may be configured to discretize the data identifying propagation surface **60** and/or any previous layer into a plurality of points, and to offset each point in propagation direction **70**. Alternatively, the data identifying propagation surface **60** may be discretized when it is received.

(98) Structure layers **72** may be determined until propagation boundary (e.g., surface **62B**) is reached. After the propagation boundary is reached, no more structure layers **72** are determined. In specific arrangements, structure layers **72** may be determined until a part of the boundary furthest from the propagation surface **60** in the propagation direction **70** is reached. Put another way, structure layers **72** may be determined until the volume defined by boundary **62** is filled.

(99) One or more structure layers **72** may be truncated or cut away. For example, referring to structure layers **72**, they are offset from the previous layer until the penultimate structure layer **72C**, which is the first structure layer to reach the surface **62B**. A part of penultimate structure layer **72C** is truncated so as to conform to surface **62B**. However, the volume defined by boundary **62** is not yet filled. Therefore, a final structure layer **72D** is offset from the penultimate structure layer **72C**. Final structure layer **72D** is truncated to conform to surface **62B** and fills the volume defined by boundary **62**. Therefore, no additional structure layers **72** are determined.

(100) In arrangements where individual points of a new layer are offset from propagation surface **60** or a previous layer to form each new layer, each new layer may have a contour that is similar (e.g., nearly identical) to propagation surface **60** or the previous layer and contained within the boundary **62** defined by the mesh (e.g., within the side walls of the mesh) at the height of each layer. It will be appreciated that any promontory or concave feature of propagation surface **60** or a previous layer will be scaled according to the size of the offset. Further, any depression or concave feature of propagation surface **60** or a previous layer will be shallowed according to the size of the offset, eventually being smoothed out over a number of layers. It should be noted that although boundary **62** is of uniform lateral dimensions at any point along the propagation direction **70**, in other arrangements the lateral dimensions of a boundary may shrink or grow as structure **12**

progresses along the propagation direction **70**, causing a corresponding area of each structure layer **72** to shrink or grow proportionally.

(101) As with base layer **68**, further tool paths are generated (Step **2412**) for each structure layer **72**. The further tool paths may be generated using any method described herein, in whole or in part, or using any method known in the art.

(102) In exemplary arrangements, propagation surface **60** or any previous layer may be discretized by mapping the 3-dimensional surface **60** or layer into a 2-dimensional space having defined points. The points lying within the boundary of the two-dimensional propagation surface **60** or previous layer may then be mapped back into the 3-dimensional space. The 2-dimensional space may be associated with a default planar surface or, as will be explained in more detail below with regard to FIGS. **20-23**, the 2-dimensional space may be customized by the user.

(103) The tool paths may be generated utilizing the points discussed above. For example, after base layer **68** or structure layer **72** has been mapped to 2-dimensions, a 2-dimensional tool path may be generated within the 2-dimensional mapping space by sequencing the points. The 2-dimensional tool path may then be mapped onto the 3-dimensional structure layer **68**, **72** using a suitable surface parameterization technique (e.g., UV parameterization, as discussed above), while maintaining the sequence of the points. In other words, each structure layer **68**, **72** of structure **12** may be mapped into the 2-dimensional space, the points truncated based on the boundary as it relates to the layer, a base tool path or further tool path generated within that space using any method described herein or an alternative method known in the art, and then mapped back to the 3-dimensional space of the corresponding layer. The user may have the ability to upload an existing tool path, sketch a tool path, pick from any number of predefined tool paths, orient the tool paths, add/remove points from a tool path, smooth a tool path **51**, and/or make other similar adjustments.

(104) As shown in FIGS. **14** and **15**, multiple different boundaries (e.g., boundaries **64**, **66**) may be used to define layers within structure **12**. For example, boundaries **62**, **64**, **66** placed at different locations and/or orientations relative to each other. As shown in FIG. **14**, surface **62B** may be defined as an additional propagation surface, and a second boundary **64** may be projected toward surface **62B** to define a second enclosed volume in which material should be deposited. A second set of structure layers, beginning with a second base layer **74** and then further structure layers, may be determined that are offset from surface **62B** and contained within sidewalls of second boundary **64**, in a similar way to that described above, and a second set of tool paths generated. It should be noted that, instead of being positioned on top of a final layer in a stack of layers propagated from surface **60** within boundary **62**, the second boundary **64** could alternatively or additionally be located adjacent a side of the stack of structure layers **72** (i.e., without regard to gravity), if desired.

(105) It should be noted that propagation of the layers within structure **12** may not be limited to a single direction. For example, had the second boundary **64** been stacked adjacent a side of the structure layers **72**, the new propagation direction could have been orthogonal to propagation direction **70**.

(106) In another example, and as also shown in FIGS. **14** and **15**, the second boundary **64** may be nested within a third boundary **66**. In this example, a third propagation surface may be defined to include part of surface **62B** and additional surface(s) formed from the second set of further layers within boundary **64**. As discussed above, the propagation direction may be any direction normal to the propagation surface at each point within the propagation surface. Accordingly, during propagation of a third set of layers from the third propagation surface, propagation may be in multiple directions at the same time (e.g., in the direction normal to surface **64B** and in an orthogonal direction—see FIG. **15**). This propagation may continue in all directions until boundary **66** is reached and the volume formed by boundary **66** is filled.

(107) When one or more boundaries (e.g., boundary **64**) are nested within another boundary **66**, a user may be able to define which boundary functions as the propagation surface and which boundary limits the propagation extent. For example, the second boundary **64** may be used alone to

generate base layer **74** and to facilitate propagation of the structure layers in the given direction up to the limit of the corresponding boundary, and thereafter the third boundary **66** may be utilized for further propagation of additional layers around the second boundary **64**. This is illustrated in FIG. **15**. However, the user could instead choose to allow the third boundary **66** to define a contour of second base layer **74**, with the second boundary **64** functioning only to truncate an area of that contour (i.e., to limit propagation of the layers outward to a volume confined by the side walls and ceiling of boundary **64**—see FIG. **16**).

(108) Similarly, the user may be able to define an order in which boundary **62** is to be used for propagation. For example, the user may be able to propagate layers within the third boundary **66** first (e.g., within a space defined outside of boundary **64**) using the contour of intersection between boundary **66** and the original boundary **62**, and thereafter to propagate layers within the second boundary **64** using the contours of intersection between boundary **64** and both of the original boundary **62** and the third boundary **66**. As illustrated in FIG. **16**, the order in which the boundaries are utilized for propagation may affect a size, shape, and contour of the resulting layers.

(109) In another example illustrated in FIGS. **20-23**, the user may be able to define one or more parameterization boundaries to be used in the mapping of points and tool paths **51** discussed above. That is, instead of always using a default parameterization boundary (e.g., the predefined points associated with build platform **32**), the user may be able to define a different and/or custom surface to function as the parameterization boundary. FIG. **20** shows build platform **32** with a first boundary **80** projected towards and, in this case intersecting surface **60** of build platform **32**. A second boundary **82** is projected towards a surface **80A** of first boundary **80**. In the exemplary arrangement of FIG. **20**, surface **80A** is a partial ellipsoid. A parameterization boundary **84** is positioned to encompass second boundary **82** and is projected toward surface **80A**.

(110) As shown in the example of FIG. **21**, using the method described above in relation to FIGS. **14-19**, layers of structure **12** are propagated through second boundary **82** by offsetting from surface **80A**. An upper surface **82A** of second boundary **82** has varying height and, in the example of FIG. **20**, is wavy. Therefore, after generating layers of structure **12** based on an intersection of a layer offset from surface **80A** with second boundary **82**, two separate three-dimensional regions **86A**, **86B** of the layer may be generated.

(111) Each of regions **86A**, **86B** may be mapped from 3-dimensions to 2-dimensional space (i.e., to correspond to build platform **32**) using a surface parameterization technique, such as UV parameterization. This process may be done before tool paths **51** are generated. Separately mapping each region **86A**, **86B** to 2-dimensions may result in each region being parameterized differently and having its own coordinate system, as shown in FIG. **21**. Due to potential errors in the parameterization technique, regions **86A**, **86B** may become misaligned and/or a spatial relationship between them may change. This may result in tool paths **51** generated for regions **86A**, **86B** being unrelated to underlying tool paths **51** (e.g., where only a single region was generated) and therefore misaligned and/or wrongly positioned.

(112) As shown in FIGS. **20**, **22** and **23**, parameterization boundary **84** may be defined to function as a mapping/parameterization space for the layers created within second boundary **82** (e.g., by intersecting second boundary **82** with layers offset from surface **80A** of first boundary **80** and truncated outlying points). As shown in FIG. **23**, a parameterization layer is taken to be the intersection of the offset layer with parameterization boundary **84**. The parameterization layer includes regions **86A**, **86B**. The entire parameterization layer may be parameterized together using parameterization boundary **84** to provide the coordinate system. By configuring the parameterization technique to preserve the spatial relationship between features of the layer, the spatial relationship between regions **86A** and **86B** is maintained. In this manner, the separate regions **86A**, **86B** may be commonly parameterized within the same space and thereafter truncated based on boundaries of the regions **86A**, **86B** at each layer, allowing correlation between the subsequently generated tool paths **51** of each region **86A**, **86B**.

(113) In exemplary arrangements, a plurality of parameterization layers may provide a common coordinate system for multiple layers of structure **12**. Referring to FIG. **15**, base layer **68** and a plurality of further layers **72** may be identical. Accordingly, the tool paths **51** for these layers may be generated more efficiently by re-using the tool path of the previous layer (e.g., via the organizational tree of FIG. **12**). This removes the computational burden of mapping each layer to 2-dimensional space and generating the tool path. The parameterization boundary may be determined such that it provides a plurality of parameterization layers having the same area when the propagation surface is offset. The parameterization layers may therefore provide a consistent coordinate system for identical layers of structure **12**. Each identical layer may be positioned at the same location within the consistent coordinate system of the parameterization layer. It is noted that this method may also be used when structure layers are similar, as defined below with respect to FIGS. **28-30**. That is, the method may be used where structure layers are not identical but have a correlation coefficient greater than a correlation threshold value.

(114) FIG. **26** shows a flow diagram for a method of determining a tool path **51**. FIGS. **25a-f** show schematic representations of a virtual model of structure **12** at different points through the method of FIG. **26**.

(115) Boundary data **2500** is received by computing device **16** (Step **2600**) and may include polygonal boundary data and/or spline boundary data. Boundary data **2500** defines, at least partially, an outer boundary **2502** of a layer of structure **12**. In the exemplary arrangement shown in FIGS. **25a-f**, boundary data **2500** is polygonal and includes a plurality of points **2504** that may be connected to define outer boundary **2502**. For clarity, FIGS. **25b-f** show an enlarged section **2506** of the virtual model of structure **12** shown in FIG. **25a**.

(116) Based on boundary data **2500**, a graph structure is determined (Step **2602**). In the exemplary arrangement of FIG. **25c**, a graph structure **2507** includes a plurality of edges **2508a-d** and a node **2512**. The edges **2508a-d** represent elongate elements **2510a-d** of structure **12** and, in the exemplary arrangement of FIG. **25c**, are located within outer boundary **2502**. Each edge **2508a-d** may be an axis of one or more elongate elements **2510a-d** of structure **12** and specifically, each edge **2508a-d** is a medial axis of an elongate member **2510a-d**. It will be appreciated that an edge **2508a-d** may be located inside or outside outer boundary **2502** and/or may be coincident with one or more elements of outer boundary **2502**.

(117) Outer boundary **2502** includes a plurality of elongate elements **2510a-d** and, in such arrangements, graph structure **2507** may be determined to include an edge **2508a-d** for each elongate element **2510a-d** that is a medial axis of each elongate element **2510a-d**, and a node **2512** at each intersection of edges **2508a-d**. Nodes may also be located at the end of an edge. Graph structure **2507** may be used to determine a plurality of regions of outer boundary **2502** that may be used during pathing. The plurality of regions may be determined by a user. In other arrangements, the plurality of regions may be determined by computing device **16** based on one or more predetermined criteria.

(118) For example, the plurality of regions may be determined such that a branch angle (i.e., an angle between elongate elements **2510a-d** forming a region) is not less than a threshold branch angle (i.e., that the branch angle is not too tight). Alternatively or in addition, the plurality of regions may be determined such that any one region does not exceed a length threshold. In exemplary arrangements, a region does not cross itself. A region includes one or more elongate elements, as defined by graph structure **2507** and outer boundary **2502**. That is, a plurality of regions may be determined by identifying one or more edges **2508a-d** of graph structure **2507** based on a walking strategy, and outer boundary **2502** of the associated elongate elements **2510a-d**. In exemplary arrangements a relationship between edges **2508a-d** and outer boundary **2502** of the associated elongate elements **2510a-d** is known such that regional boundaries may be formed from edges **2508a-d**.

(119) FIG. **25d** shows three regions **2514a-c**. A first region **2514a** includes elongate elements

**2510b** and **2510c**, a second region **2514b** includes elongate element **2510a** and a third region **2514c** includes elongate element **2510d**. Using the method described above, regions of a layer of structure **12** may be determined easily and in a way that allows them to be pathed separately.

(120) Once regions **2514a-c** have been determined, tool paths **51** may be generated for each region **2514a-c** (Step **2604**). Tool paths **51** may be generated using any process disclosed herein.

(121) As seen in FIG. **25e**, computing device **16** may, for example, generate a plurality of adjacent tool paths that are each offset from an edge **2508a-d** of graph structure **2507**. This is shown in the case of first region **2514a** formed by elongate elements **2510b**, **2510c**. FIG. **25e** shows adjacent tool paths **2516** offset from edges **2508b**, **2508c** of first region **2514a** in a direction normal to edges **2508b**, **2508c**. Although not shown in the drawings, it is noted that edges **2508a-d** may be curved or straight. Adjacent tool paths **2516** may be offset by a distance equivalent to a thickness of a material to be deposited by machine **14** when fabricating structure **12**. Although not shown in FIG. **25e**, further adjacent paths may be offset in an opposite direction from edges **2508b**, **2508c**. In addition, adjacent paths may be generated that are offset from the remaining edges **2508a**, **2508d**. This is discussed in greater detail below. It is also noted that adjacent tool paths **2516** may be offset from outer boundary **2502** or another reference.

(122) Adjacent tool paths **2516** are generated that cover an area extending beyond outer boundary **2502**, as can be seen in FIG. **25e**. Boundary data **2500** may be superimposed on adjacent paths **2516**, which may then be adapted, for example by clipping or diverting one or more adjacent paths **2516**, so that they are contained within outer boundary **2502**. In exemplary arrangements, adjacent paths for each region **2514a-c** may be generated separately. Accordingly, regional boundary data defining a boundary of each region **2514a-c** may be superimposed on adjacent tool paths **2516**, which may be adapted to be contained within the regional boundary data. For example, adjacent tool paths **2516** may be clipped to be contained within regional boundary lines **2518a-b** as they taper from a relatively wider elongate element **2510b** to a relatively narrower elongate element **2510c**. Adjacent tool paths for other regions **2514b**, **2514c** may be generated and/or adapted such that they do not overlap with adjacent paths **2516** of first region **2514a**. In exemplary arrangements, tool paths for second and third regions **2514b**, **2514c** may be generated separately, or as single tool paths that are then clipped by superimposing regional boundary data onto them. FIG. **25f** shows adapted tool paths within outer boundary **2502**. As can be seen in FIG. **25f**, the adjacent paths of second and third regions **2514b**, **2514c** may be terminated before they intersect with any of adjacent tool paths **2516** of first region **2514a**.

(123) Structure **12** may be formed of a plurality of overlapping layers of material, and boundary data **2500** may relate to at least part of a first layer that, when fabricated, will be overlapped by a second layer. In such arrangements, the regions **2514a-c** in the first layer may be defined such that they are not aligned with the regions in the second layer. For example, if regions **2514a-c** are defined in the second layer, then in the first layer a first region may include elongate elements **2510a** and **2510d**, a second region may include elongate element **2510b**, and a third region may include elongate element **2510c**. In this way, the boundaries of regions within overlapping layers may be misaligned to improve a property (e.g., a strength) of structure **12**. Further, adjacent tool paths of the first layer may be offset from adjacent tool paths of the second layer.

(124) In the exemplary arrangement shown in FIG. **25f**, adjacent tool paths within the same layer are offset from each other by the width of material to be deposited by machine **14** during fabrication. However, in other arrangements adjacent tool paths may be offset by a greater amount. In yet further arrangements part of the adjacent tool paths may be offset from each other by the width of material to be deposited and part of the adjacent tool paths may be offset by a greater amount.

(125) Outer boundary **2502** may form part of a non-planar layer of structure **12**. The non-planar layer may be mapped or transformed into 2-dimensions before the plurality of adjacent tool paths are generated. After the adjacent tool paths have been adapted, the adjacent tool paths may be



mapped back from 2-dimensions to conform to the non-planar layer. The mapping may be done using a surface parameterization technique, such as UV parameterization or any other known technique.

(126) In arrangements discussed above, adjacent tool paths are spatially offset from edges **2508a-d** of graph structure **2507** and other adjacent tool paths. It is envisaged that adjacent tool paths may be rotationally and/or spatially offset from edges **2508a-d** of graph structure **2507**, for example in a way similar to the spokes of a wheel. Further, edges **2508a-d** of graph structure **2507** may be curved or linear and, for curved edges, the offset may be in a direction normal to any point on the edge.

(127) Referring to FIG. 27, two or more regions **2514d-e** may form at least part of an elongate element **2510e**. In the example of FIG. 27, part of regions **2514d** and **2514e** run parallel with each other within elongate element **2510e**. Regions **2514d-e** may form at least part of a plurality of elongate elements. For example, region **2514e** may extend or be diverted into a transverse elongate element **2510f** and/or region **2514d** may extend into an elongate element **2510g** continuing in the same direction from elongate element **2510e**. Regions **2514d-e** may combine with further regions to form further elongate elements. Regions **2514d-e** may also recombine to form a further elongate element.

(128) In exemplary arrangements, after graph structure **2507** has been formed and the plurality of regions **2514a-c** defined, tool paths **51** may be generated using alternative methods. Tool paths **51** may be generated using any method described herein or any method known in the art. For example, tool paths **51** of regions **2514a-c** may be determined using partitioning lines, as in the method described in relation to FIGS. 3-10. After such methods an outer wall of regions **2514a-c** may be formed and infill may be provided to the regions **2514a-c** by offsetting a plurality of adjacent layers from edges **2508a-d** of graph structure **2507**. It is also noted that tool paths **51** may be generated for any region (e.g., without the need to first generate graph structure **2507**) using the method described in relation to FIGS. 3-10 combined with infill provided by offsetting a plurality of adjacent layers from a reference for the region. The reference may, for example, be a medial axis of the region, an outer boundary of the region or any other reference.

(129) The layer of structure **12** defined by boundary data **2500** may be a first layer onto which a second, overlapping layer is to be deposited. In exemplary arrangements, graph structure **2507** of the first layer may be compared to a graph structure of the second layer to determine a correlation between the two graph structures. Based on the correlation (i.e., if the correlation is above a threshold value), then the walking strategy used to determine regions **2514a-c** of the first layer may also be used, at least in part, to determine regions of the second layer. This may be advantageous in cases where a coefficient of the correlation is above a threshold value but less than 1, for example where the correlation is high but the graph structures are not identical.

(130) Referring to FIGS. 28 and 29, examples are given of layers of structure **12** having similar but non-identical graph structures. FIG. 28 shows a twisted structure **2800** comprised of a plurality of layers. Each layer is rotated with respect to the previous layer to produce twisted structure **2800**. As a result of the rotation, subsequent layers of twisted structure **2800** are not identical, even though they include the same geometric structure. FIG. 29 shows a graph structure **2900** of a first layer and a graph structure **2902** of a second layer. Graph structure **2900** includes a plurality of first nodes **2904** and first edges **2906**. In particular, graph structure **2900** includes first node **2904a** and first edge **2906a**. Graph structure **2902** includes a plurality of second nodes **2908** and second edges **2910**. The second nodes **2908** and second edges **2910** are geometrically the same (the edges have the same size and orientation, and the nodes have the same position) as the first nodes **2904** and second nodes **2906**, except that there is no second node corresponding to first node **2904a** and no second edge corresponding to first edge **2906a**. Despite this, the walking strategy for graph structure **2900** may be used for graph structure **2902**, as it relates to the corresponding nodes and edges.

(131) FIG. 30 is a flowchart showing a method of determining a tool path of a second layer overlapping a first layer. In a manner discussed above, first graph structure **2900** is determined based on boundary data **2500** and a plurality of regions and the tool paths are determined for the first layer (Step **3000**). Second boundary data is received that at least partially defines an outer boundary of a second layer of structure **12** that will overlap the first layer. Based on the second boundary data, second graph structure **2902** for the second layer is determined (Step **3002**). This may be done using a similar method to that discussed above in respect of the first layer.

(132) First and second graph structures **2900**, **2902** may be compared to determine a correlation between them (Step **3004**). Determining the correlation may include determining a geometric similarity between first and second graph structures **2900**, **2902**. Geometric similarity need not be based on a position and/or orientation of the second layer with respect to the first layer when deposited by machine **14**. Further, geometric similarity need not be based on an overall scale of the second layer with respect to the first layer (i.e., consecutive layers may be scaled versions of each other). Instead, geometric similarity may relate to the relative positions between first and second nodes **2904**, **2908** and edges **2906**, **2910** within each graph structure **2900**, **2902**. That is, correlation and geometric similarity may be based on the size and shape of each graph structure **2900**, **2902** and not on the position and/or orientation of each layer within structure **12**.

(133) One or more first nodes **2904** may be determined to be correlated with corresponding second nodes **2908** if they are geometrically positioned within a distance threshold of the corresponding second nodes **2908**. One or more first edges **2906** may be determined to be correlated with corresponding second edges **2910** if they have an angle within an angle threshold and/or a length within a length threshold of the corresponding second edges **2908**. Correlation may be concluded based on a node-by-node or edge-by-edge comparison or based on a comparison of plurality of first nodes and/or edges to a plurality of a second nodes and/or edges. A correlation coefficient may be assigned to each comparison. If the correlation coefficient is greater than a coefficient threshold (or the graph structure **2900**, **2902** are otherwise determined to be correlated) then the regions of the second layer may be determined based on the walking strategy used in determining the regions of the first layer. Optionally, tool paths of the second layer may be determined based on the tool paths of the first layer. In exemplary arrangements, the correlation coefficient may be less than 1 for graph structures **2900**, **2902**, because they are not identical.

(134) In exemplary arrangements, a weighting may be applied to one or more of first and/or second nodes **2904**, **2908** and/or to first and/or second edges **2906**, **2910**. The weighting may signify an importance that a first node **2904** or first edge **2906** is correlated with a corresponding second node **2908** or second edge **2910**. The weighting may be used in determining the overall correlation.

(135) It will be apparent to those skilled in the art that various modifications and variations can be made to the disclosed systems and methods. Other embodiments will be apparent to those skilled in the art from consideration of the specification and practice of the disclosed systems and methods. It is intended that the specification and examples be considered as exemplary only, with a true scope being indicated by the following claims and their equivalents.

## Claims

1. A method of determining a tool path for use in additively manufacturing a structure, the method comprising: receiving data that at least partially defines an outer boundary for a region of the structure; partitioning the region into a first partition and a second partition based on placement of a partitioning line across the region from the outer boundary; generating the tool path to navigate at least part of the outer boundary and to cross, at least partially, the region from the first partition into the second partition based on the placement of the partitioning line; and controlling an additive manufacturing machine to manufacture the structure based on the tool path; and generating the tool path to navigate at least part of the outer boundary and to cross through the region includes

generating the tool path to cross over itself at an intersection within a first layer of the structure; and legs of the intersection are laterally shifted relative to a second layer of the structure, such that a thickness of the intersection of the first and second layers is the same as a combined thickness of first and second layers.

2. The method of claim 1, wherein the placement of the partitioning line is determined based on a desired property for the structure.
3. The method of claim 2, wherein the desired property includes a load the structure is designed to withstand.
4. The method of claim 3, further including modeling behavior of the structure under a given condition, and determining the desired property based on the modeling.
5. The method of claim 4, wherein modeling includes generating a property vector field through the structure.
6. The method of claim 4, wherein modeling includes performing a finite element analysis of the region.
7. The method of claim 1, further comprising: modeling behavior of the structure during use based on the tool path; adjusting the partitioning line based on the modeling; and regenerating the tool path based on the adjusted partitioning line.
8. The method of claim 1, wherein: the partitioning line is a first partitioning line; the method further includes partitioning the region based on placement of a second partitioning line extending across the region from the outer boundary to form an intersection lane therebetween; and the tool path crosses through the region within the intersection lane.
9. The method of claim 1, wherein generating the tool path includes: generating a section of the tool path navigating along a part of the outer boundary; and the method further includes: when the partitioning line is encountered, diverting the tool path to cross through the region; and when a further part of the outer boundary is encountered, diverting the tool path to navigate along the further part of the outer boundary.
10. The method of claim 9, wherein: the tool path has a current direction of propagation; diverting the tool path to cross through the region includes diverting in a direction that is one of left or right of the current direction of propagation; and diverting the tool path to navigate along the further part of the outer boundary includes diverting in a direction that is the other of left or right of the current direction of propagation.
11. The method of claim 1, further including, prior to partitioning the region, generating a seam that is internal to the region, wherein partitioning the region includes placing the partitioning line to extend from the outer boundary to the seam.
12. The method of claim 11, wherein the tool path further navigates along the seam.
13. The method of claim 1, further including generating an outer tool path that circumnavigates of the outer boundary and that does not cross through the region.
14. The method of claim 13, further including combining the tool path and the outer tool path within a single layer of the structure.
15. The method of claim 13, further including: using the tool path alone to fabricate a first layer of the structure; and using the outer tool path alone to fabricate a second layer of the structure.
16. The method of claim 1, further including compiling and outputting machine code including instructions for controlling an additive manufacturing machine based on the tool path.
17. A non-transitory computer-readable medium encoded with a computer program that, when executed on a computer processor, is configured to undertake steps of: receiving data that at least partially defines an outer boundary for a region of the structure; partitioning the region into a first partition and a second partition based on placement of a partitioning line across the region from the outer boundary; generating the tool path to navigate at least part of the outer boundary and to cross, at least partially, the region from the first partition into the second partition based on the placement of the partitioning line; and controlling an additive manufacturing machine to manufacture the

structure based on the tool path; and generating the tool path to navigate at least part of the outer boundary and to cross through the region includes generating the tool path to cross over itself at an intersection within a first layer of the structure; and legs of the intersection are laterally shifted relative to a second layer of the structure, such that a thickness of the intersection of the first and second layers is the same as a combined thickness of first and second layers.

18. An additive manufacture system, comprising: a moveable support; a print head connected to the moveable support and configured to discharge a material; and a computer processor including computer program code that, when executed on the computer processor, is configured to undertake steps of: receiving data that at least partially defines an outer boundary for a region of the structure; partitioning the region into a first partition and a second partition based on placement of a partitioning line across the region from the outer boundary; generating the tool path to navigate at least part of the outer boundary and to cross, at least partially, the region from the first partition into the second partition based on the placement of the partitioning line; and controlling an additive manufacturing machine to manufacture the structure based on the tool path; and generating the tool path to navigate at least part of the outer boundary and to cross through the region includes generating the tool path to cross over itself at an intersection within a first layer of the structure; and legs of the intersection are laterally shifted relative to a second layer of the structure, such that a thickness of the intersection of the first and second layers is the same as a combined thickness of first and second layers.

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